

Federated Biobank Analytics

Agent orchestration across three data sites

NVFlare for distributed execution

Codex for analysis planning and development

Hackathon prototype

Human analysis compared with agent-generated results



The analysis problem

A researcher wants one answer from several biobanks, but the records cannot move to a central database.

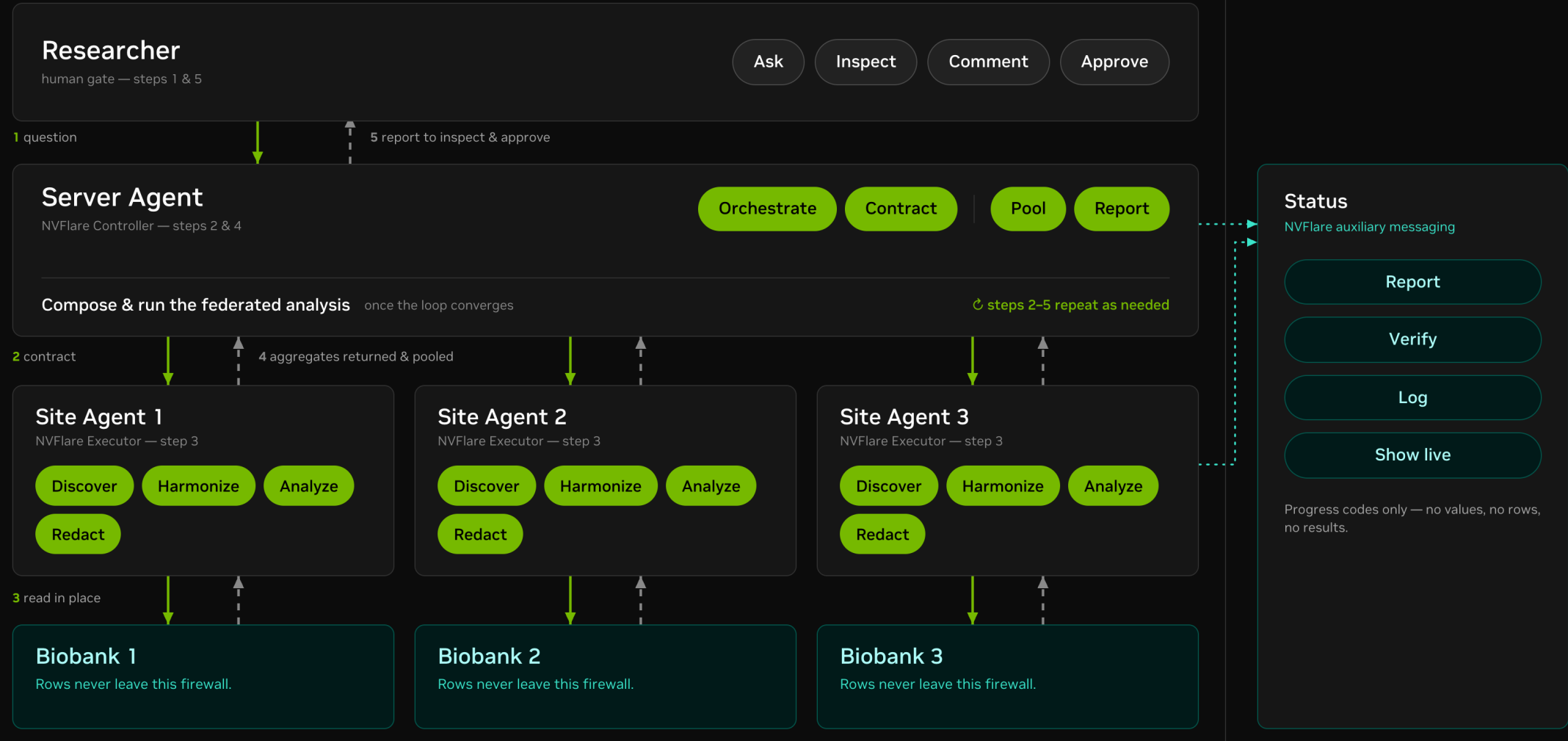
- 01 Different schemas** Equivalent concepts may use different column names, units, or encodings.
- 02 Protected data** Patient-level rows must remain inside each institution.
- 03 Multi-site statistics** Site results require mathematically correct pooling, not simple averaging.

Orchestrated end to end by NVFlare

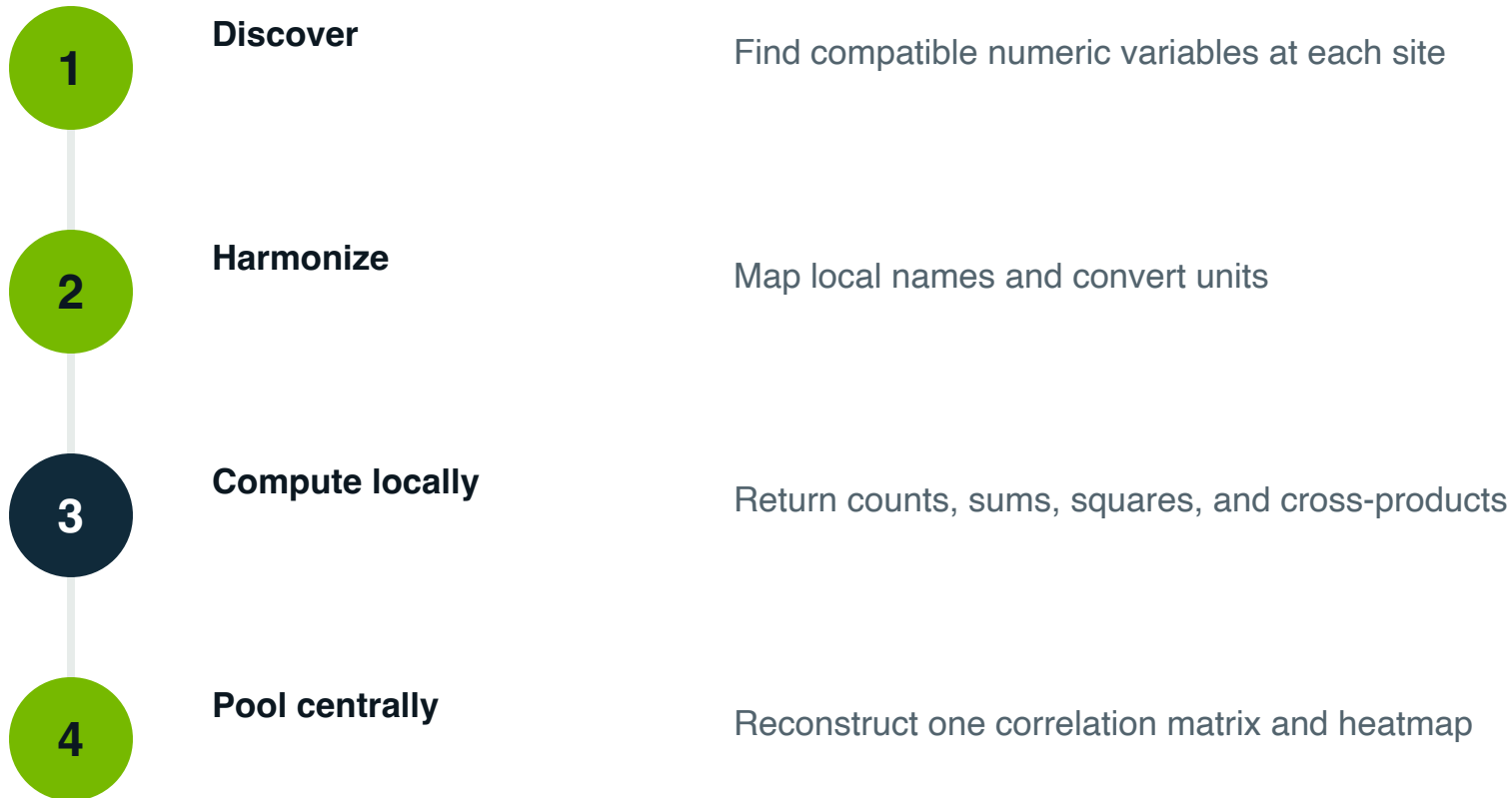
Server agent = Controller. Site agents = Executors. Data stays home. Two channels, one job: the analysis flow on the left, the UI status stream on the right.

MAIN FLOW — ORCHESTRATED BY NVFLARE

UI STATUS — NVFLARE MESSAGING



Federated correlation workflow



Human and agent correlation comparison - Site A

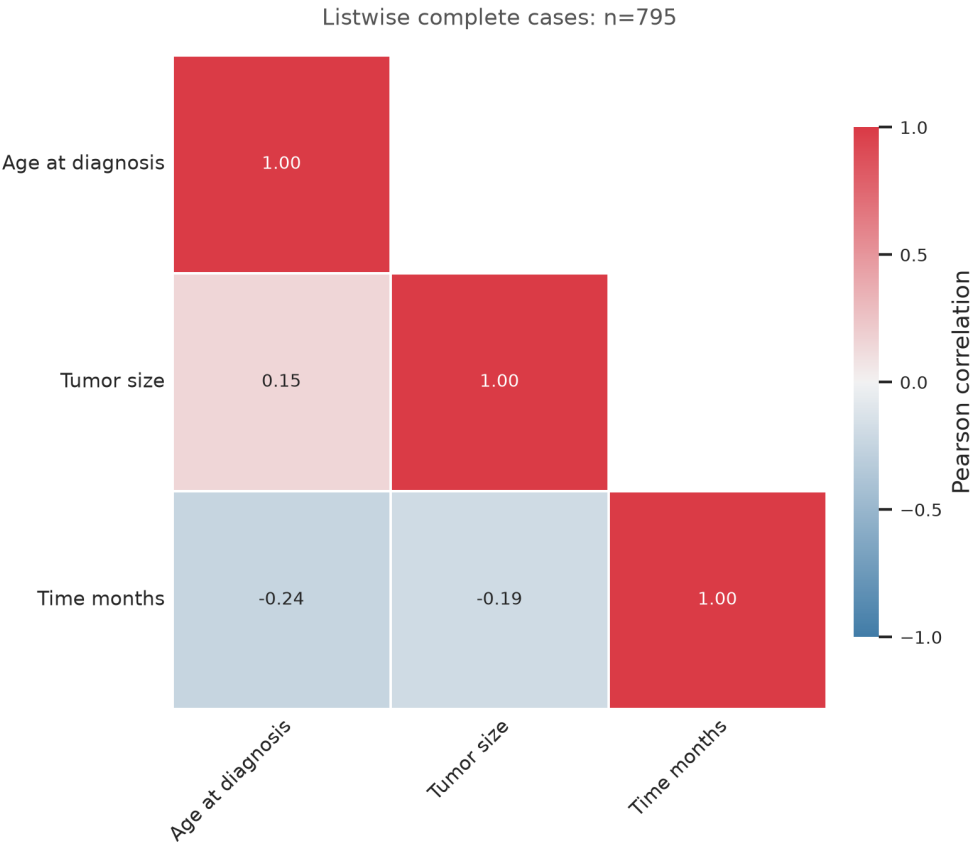
Agent analysis

Site And Federated Pearson — Site A University Hospital



Human analysis

University Hospital: Local correlation heatmap



Values show Pearson correlation. Blank cells indicate undefined correlation, usually caused by a constant variable.

Human and agent correlation comparison - Site B

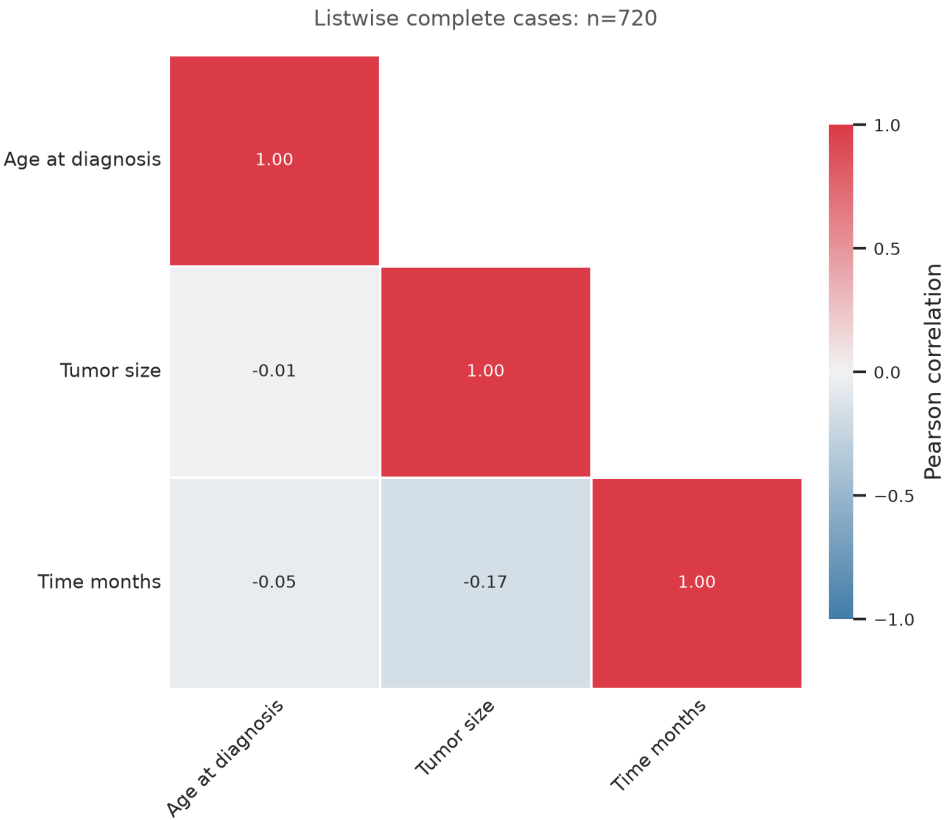
Agent analysis

Site And Federated Pearson — Site B Cancer Centre



Human analysis

Cancer Centre: Local correlation heatmap



Values show Pearson correlation. Blank cells indicate undefined correlation, usually caused by a constant variable.

Human and agent correlation comparison - Site C

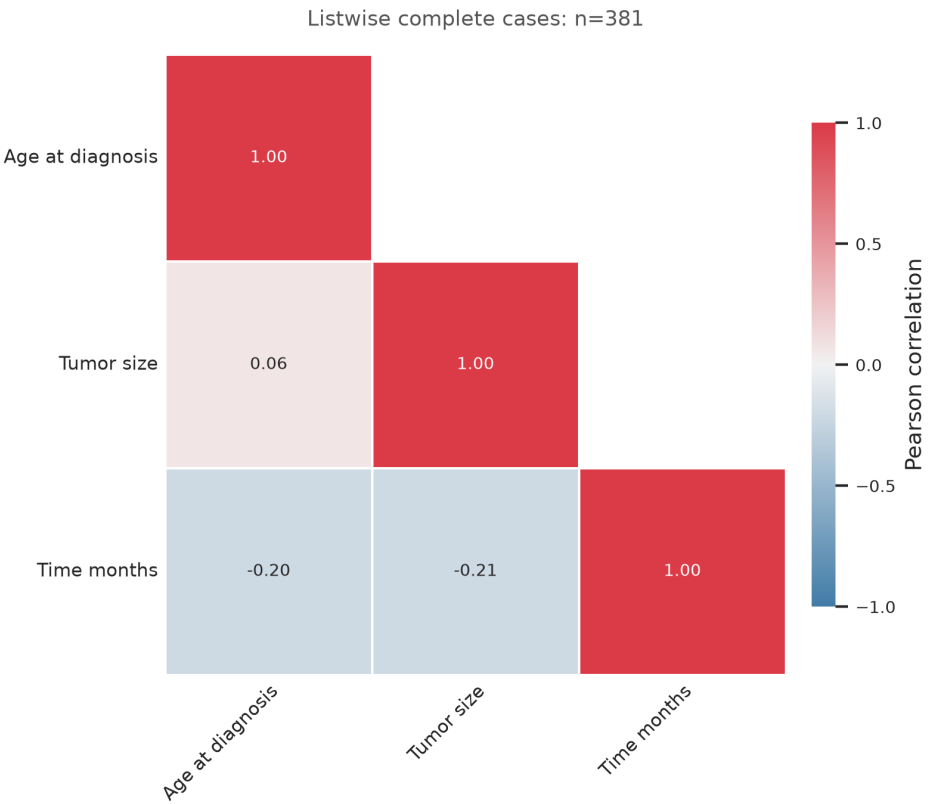
Agent analysis

Site And Federated Pearson — Site C Regional Hos



Human analysis

Regional Hospital: Local correlation heatmap



Values show Pearson correlation. Blank cells indicate undefined correlation, usually caused by a constant variable.

Human and agent correlation comparison - Federated

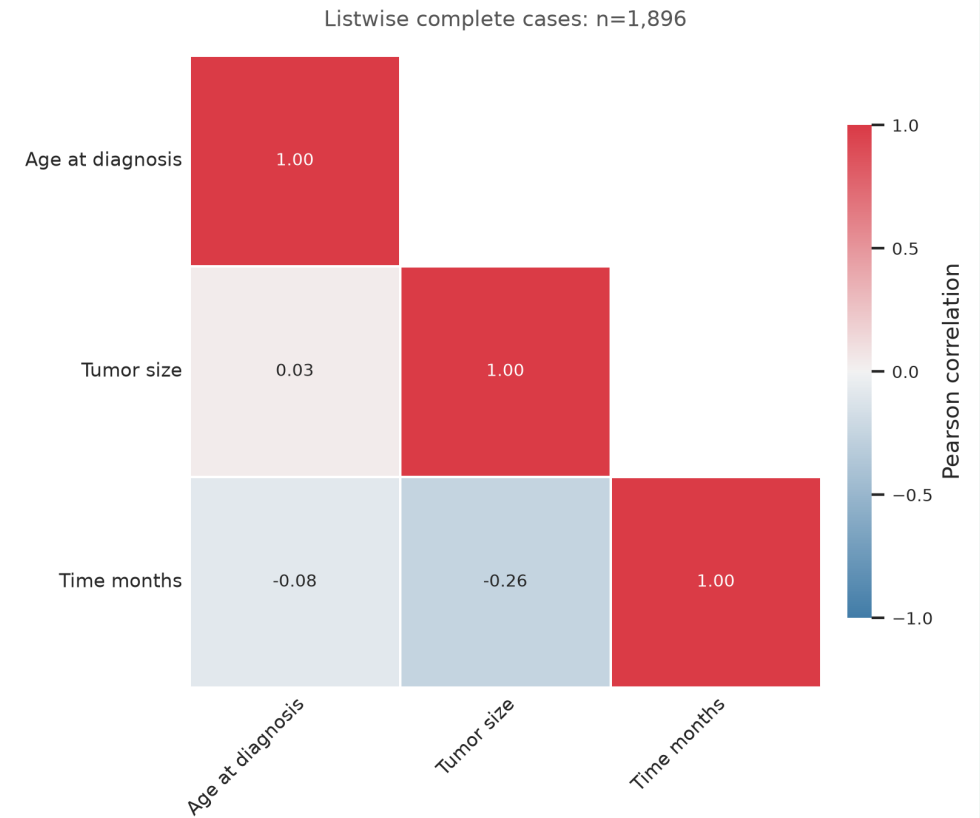
Agent analysis

Site And Federated Pearson — Federated

	age_at_diagnosis	tumor_size	time_months
age_at_diagnosis	1.000	-0.033	-0.146
tumor_size	-0.033	1.000	-0.141
time_months	-0.146	-0.141	1.000

Human analysis

Federated METABRIC correlation heatmap



Values show Pearson correlation. Blank cells indicate undefined correlation, usually caused by a constant variable.

What the prototype demonstrates

- 01 Ask in natural language**

The agent converts an analysis question into a structured, inspectable plan.
- 02 Execute under governance**

NVFlare sends approved tasks to local sites while patient records remain local.
- 03 Reproduce the result**

Human and agent outputs can be compared with explicit numerical tolerances.